

Consumer's equilibrium

Consumer = Max Satisfied

Consume $\rightarrow$ unchanged

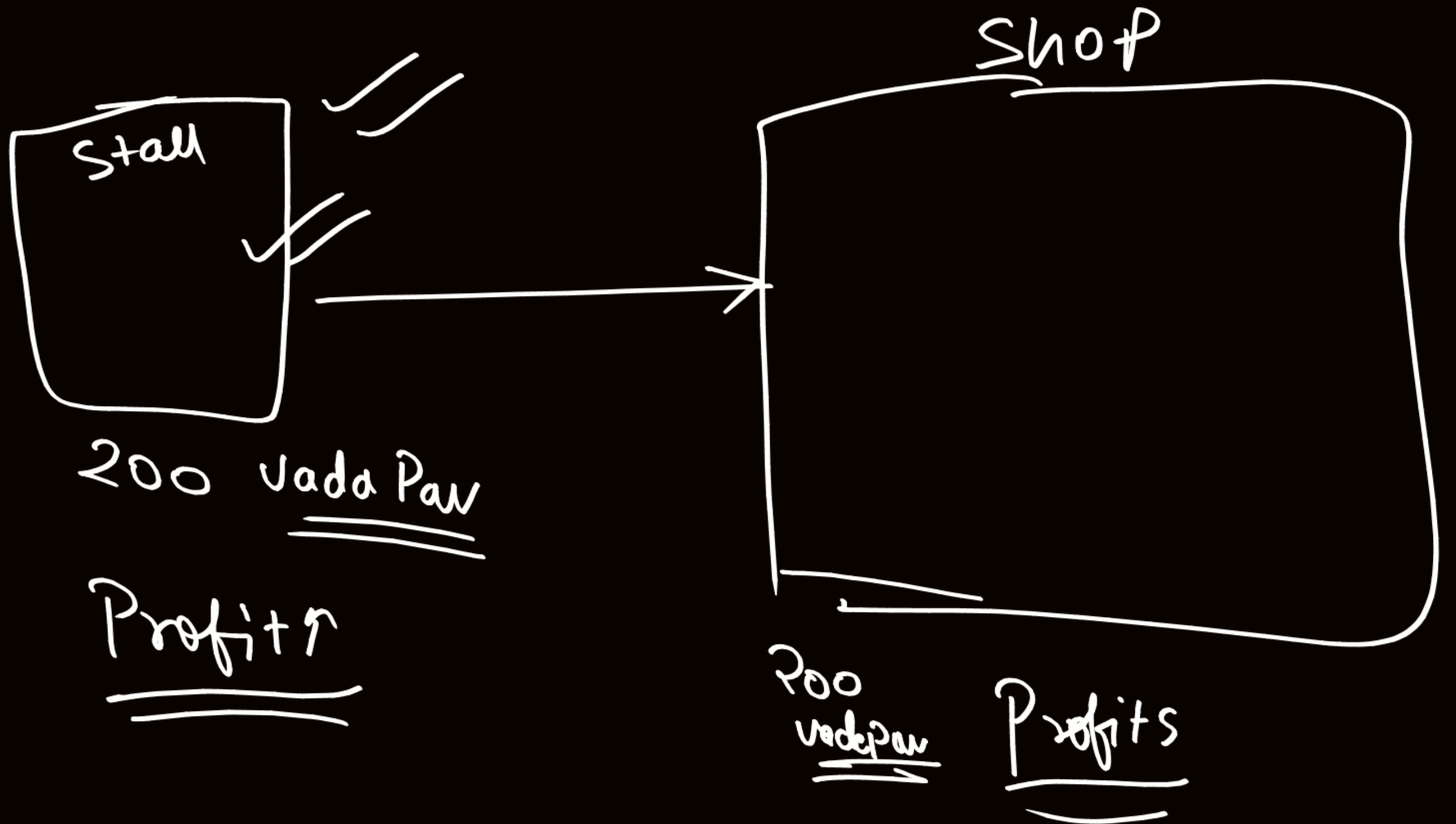
① C.E. $P_x = MU_x$

② $MU_x = MU_y \rightarrow$ C.E.



MICROECONOMICS
PRODUCER'S
EQUILIBIRUM

Chapter 8 | Lecture 1



MEANING OF PROFIT

Profit refers to the excess of receipts from the sale of goods over the expenditure incurred on producing them.

Cost
₹ 28

Revenue
₹ 10



₹ 22

Revenue



Expenditure



Milk
Water
Sugar
Tea
Ginger
elaichi
Chai
Masala

Profit

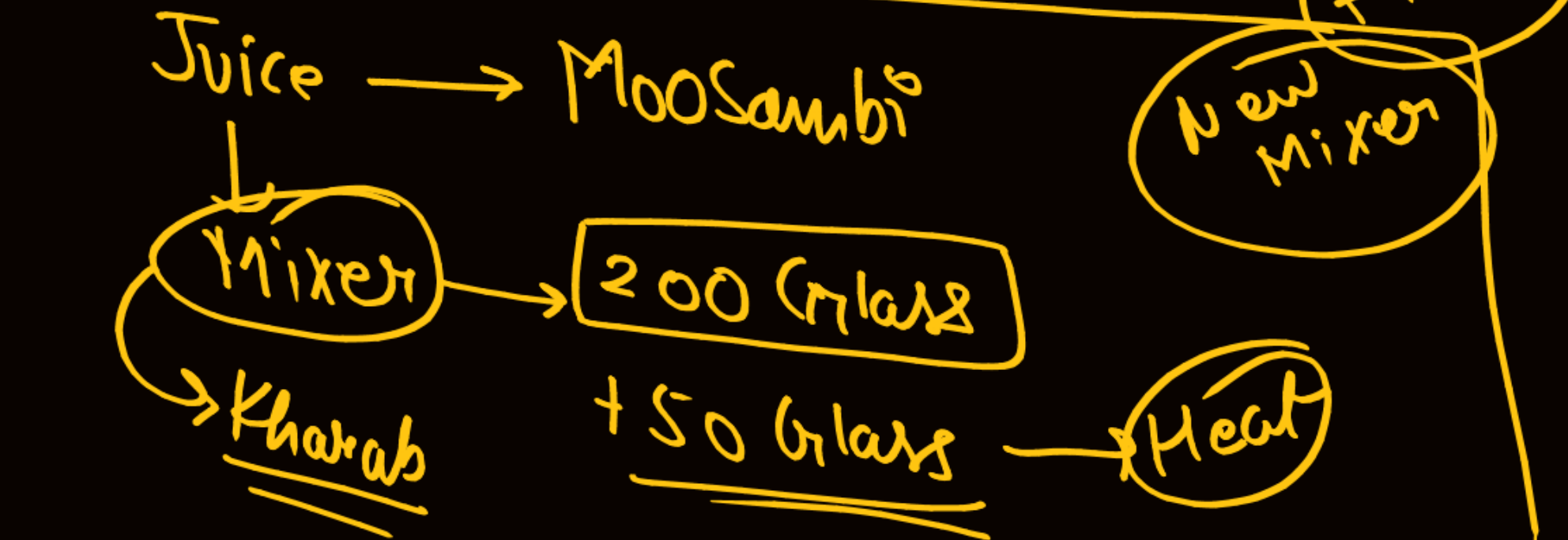


PRODUCER'S EQUILIBRIUM

Producer's Equilibrium refers to that price and output combination which brings maximum profit to the producer and profit declines as more is produced.



1
2
3
4
5
6
7
8
9
10 → 11
x 11 +



PRODUCER'S EQUILIBRIUM

There are two methods for determination of producer's equilibrium:

**Total Revenue and
Total Cost Approach
(TR-TC Approach)**



**Marginal Revenue &
Marginal Cost Approach
(MR-MC Approach)**

TR-TC Approach

Output	TR	TC	Profit (TR-TC)
0	0	10	-10
1	20	15	5
2	40	25	15
3	60	35	25
4	80	60	20
5	100	80	20

P.E

Max. Profit

$$TC = TFC + TVC$$
$$10 = 10 \quad \bigcirc$$

PRODUCER'S EQUILIBRIUM

Producer can attain the equilibrium level under two different situations:

Perfect competition

When Price Remains Constant

(It happens under Perfect Competition). In this situation, firm has to accept the same price as determined by the industry. It means, any quantity of a commodity can be sold at that particular price.



Imperfect competition

When Price Falls With Rise in Output

(It happens under Imperfect Competition). In this situation, firm follows its own pricing policy. However, it can increase sales only by reducing the price.

MC - MR Approach

→ ① $MC = MR$ //

1 10 14

2 10 12

3 10 10 -

→ ② $MC > MR$ //

Producer's equilibrium

$MC = +1 \rightarrow$ Banane ki Cost

$MR = +1 \rightarrow$ Product ko Sell karke kitna paisa kamaya

MC - MR Approach

Output	MC	MR
1	4	10
2	6	10
3	8	10
4	10	10
5	6	10
6	10	10
7	12	10

① ✓

X MC = MR ✓

① ✓ MC = MR

②

MC > MR

P.E

X

MARGINAL REVENUE-MARGINAL COST APPROACH (MR-MC APPROACH)

①
MC = MR

As long as MC is less than MR, it is profitable for the producer to go on producing more because it adds to its profits. The producer stops producing more only when MC becomes equal to MR.

MC is greater than MR
after MC=MR output level

When MC is greater than MR after equilibrium, "it means producing more will lead to decline in profits."

MARGINAL REVENUE-MARGINAL COST APPROACH

(MR-MC APPROACH)

H.W

Both the conditions are needed for Producer's Equilibrium

MC = MR

We know, MR is the addition to TR from sale of one more unit of output and MC is addition to TC for increasing production by one unit. Every producer aims to maximize the total profits. For this, a firm compares its MR with its MC. Profits will increase as long as MR exceeds MC and profits will fall if MR is less than MC. So, equilibrium is not achieved when $MC < MR$ as it is possible to add to profits by producing more. Producer is also not in equilibrium when $MC > MR$ because benefit is less than the cost. It means, the firm will be at equilibrium when $MC = MR$.

MC is greater than MR after MC=MR output level

MC = MR is a necessary condition, but not sufficient enough to ensure equilibrium. It is because MC = MR may occur at more than one level of output. However, out of these, only that output level is the equilibrium output when MC becomes greater than MR after the equilibrium. It is because if MC is greater than MR, then producing beyond MC = MR output will reduce profits. On the other hand, if MC is less than MR beyond MC = MR output, it is possible to add to profits by producing more. So, first condition must be supplemented with the second condition to attain the producer's equilibrium.

MARGINAL REVENUE-MARGINAL COST APPROACH (MR-MC APPROACH)

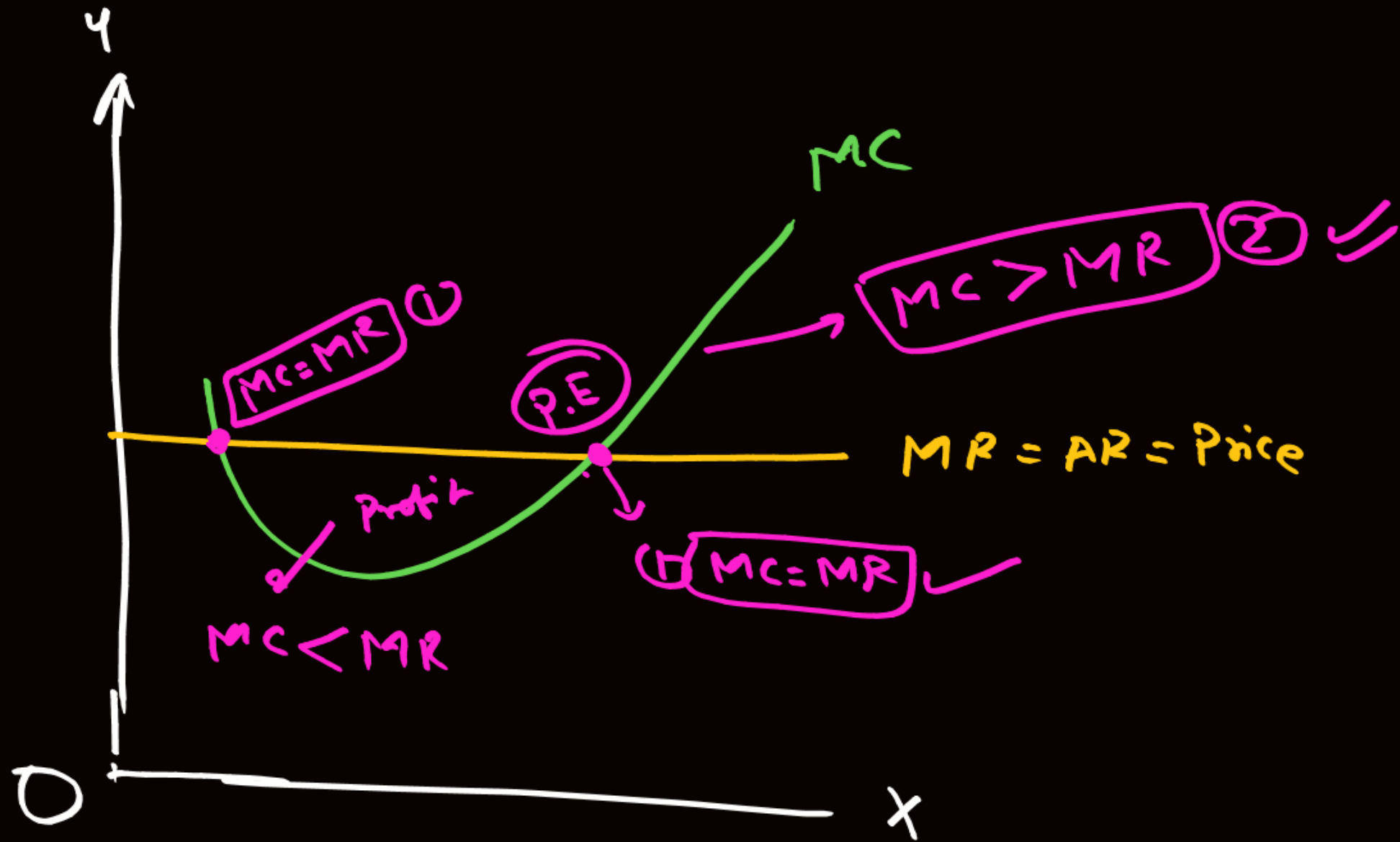
Producer's Equilibrium (When Price remains Constant)

Output (Units)	Price (₹)	TR (₹)	TC (₹)	MR (₹)	MC (₹)	Profit = TR - TC (₹)
1	12	12	13	12	13	-1
2	12	24	25	12	12	-1
3	12	36	34	12	9	2
4	12	48	42	12	8	6
5	12	60	54	12	12	6
6	12	72	68	12	14	4

$MC = MR$ (1) ✗

$MR = MC$ (1) ✓ → P.E
 $MC > MR$ (2) ✓

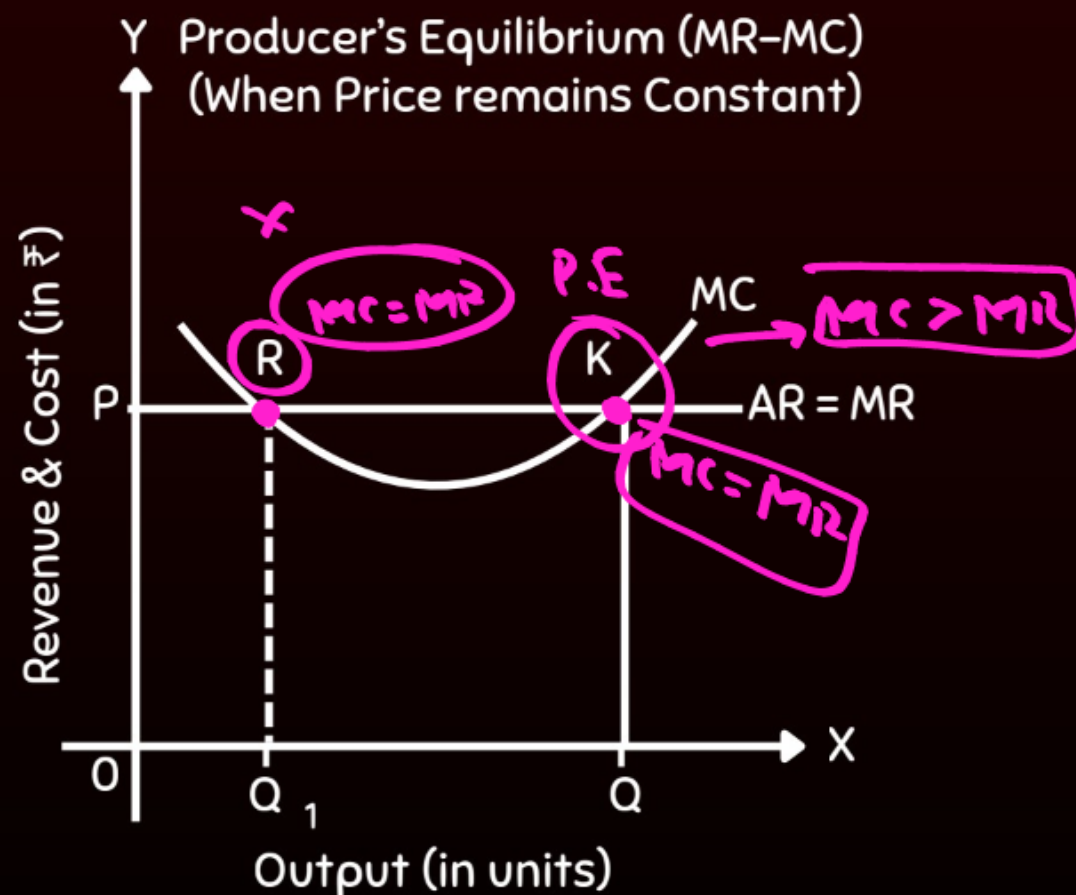
Price
constant
Perfect
Competition



MARGINAL REVENUE-MARGINAL COST APPROACH (MR-MC APPROACH)

Producer's Equilibrium (When Price remains Constant)

When price remains constant, firms can sell any quantity of output at the price fixed by the market. Producer aims to produce that level of output at which MC is equal to MR and MC is greater than MR after MC = MR output level.



- Producer's Equilibrium is determined at OQ level of output corresponding to Point K as at this point: (i) $MC = MR$; and (ii) MC is greater than MR after $MC = MR$ output level.
- Although $MC = MR$ is also satisfied at point R, but it is not the point of equilibrium as it satisfies only the first condition (i.e. $MC = MR$). So, the producer will be at equilibrium at point K when both the conditions are satisfied.

MARGINAL REVENUE-MARGINAL COST APPROACH (MR-MC APPROACH)

Relation between Price & MC at Equilibrium (WHEN PRICE FALLS WITH RISE IN OUTPUT)
Let us understand this with the help of table-

Producer's Equilibrium (When Price Falls with rise in output)

Output (Units)	Price (₹)	TR (₹)	TC (₹)	MR (₹)	MC (₹)	Profit = TR - TC (₹)
1	8	8	6	8	6	2
2	7	14	11	6	5	3
3	6	18	15	4	4	3
4	5	20	20	2	5	0
5	4	20	26	0	6	-6

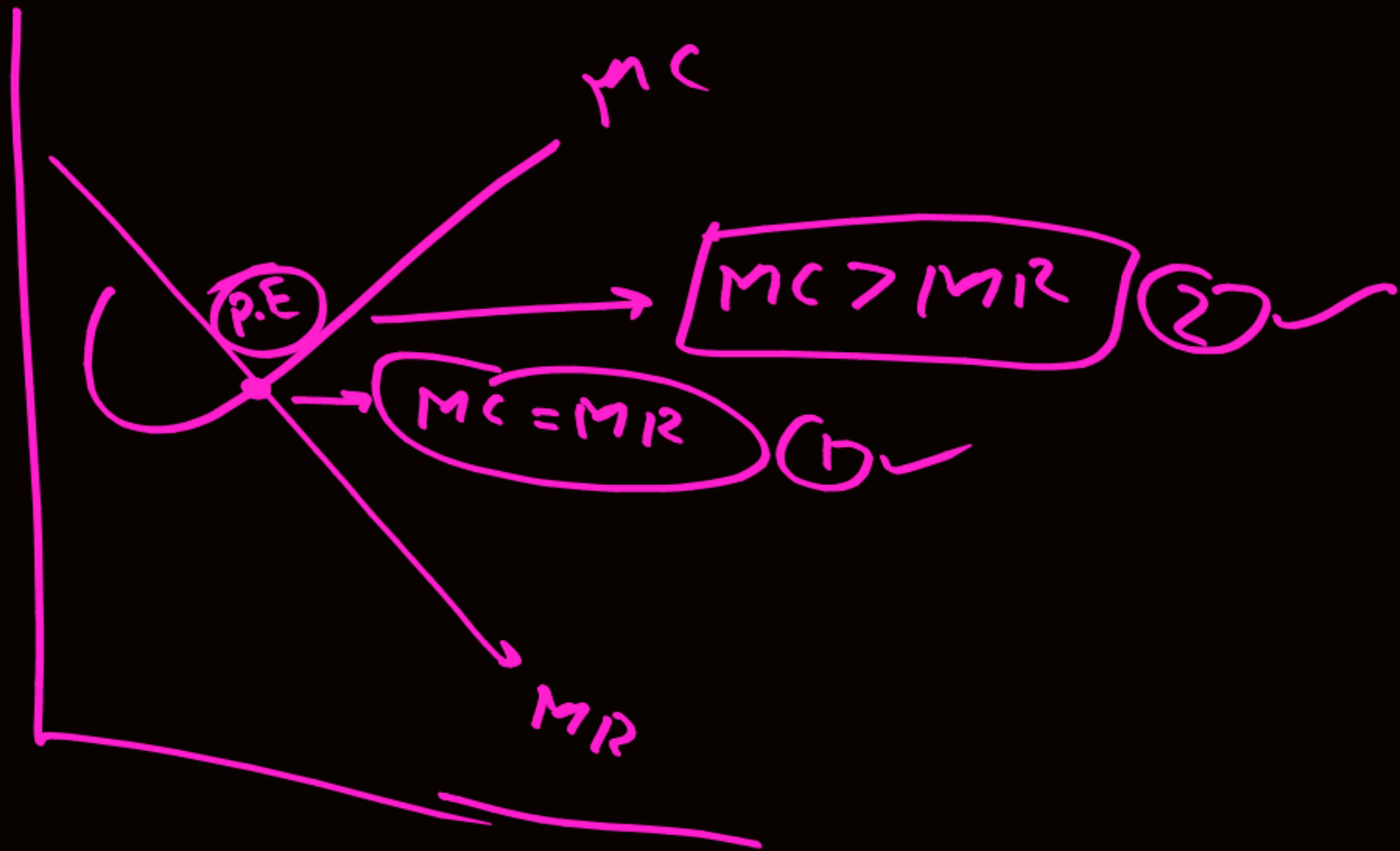
✓
TR - TC

MC = MR (1) ✓

P.E

MC > MR (2) ✓

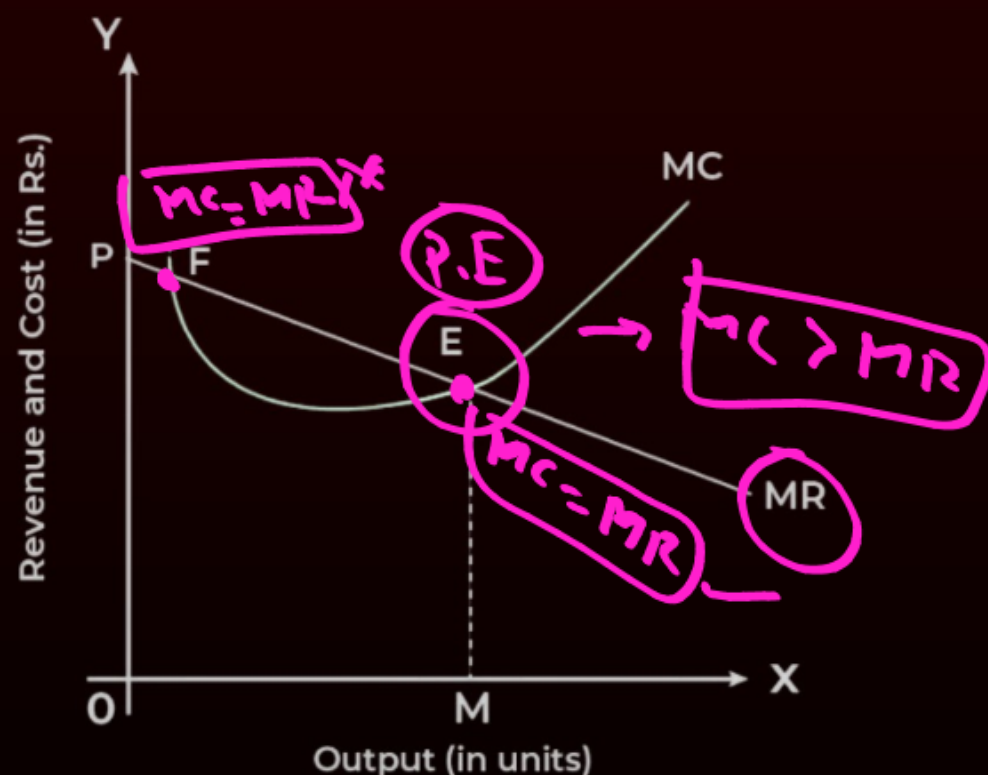
Price change



MARGINAL REVENUE-MARGINAL COST APPROACH (MR-MC APPROACH)

When there is no fixed price and price falls with rise in output, MR curve slope downwards.
Producer aims to produce that level of output at which MC is equal to MR and MC curve cuts the MR curve from below.

Producer's Equilibrium (MR-MC)
(When Price falls with rise in output)



- Producer's Equilibrium is determined at OM level of output corresponding to Point E as at this point: (i) $MC = MR$; and (ii) MC is greater than MR after $MC = MR$ output level.
- Relation between Price and MC at Equilibrium (When Price Falls with rise in output)
- When more output can be sold only by reducing the prices, then Price (or AR) $> MR$. As equilibrium is achieved when $MC = MR$, it means, price is more than MC at the equilibrium level.

MARGINAL REVENUE-MARGINAL COST APPROACH (MR-MC APPROACH)

So, the producer is at equilibrium at OM units of output.

Relation between Price and MC at Equilibrium (When Price Falls with rise in output)

When more output can be sold only by reducing the prices, then Price (or AR) > MR.

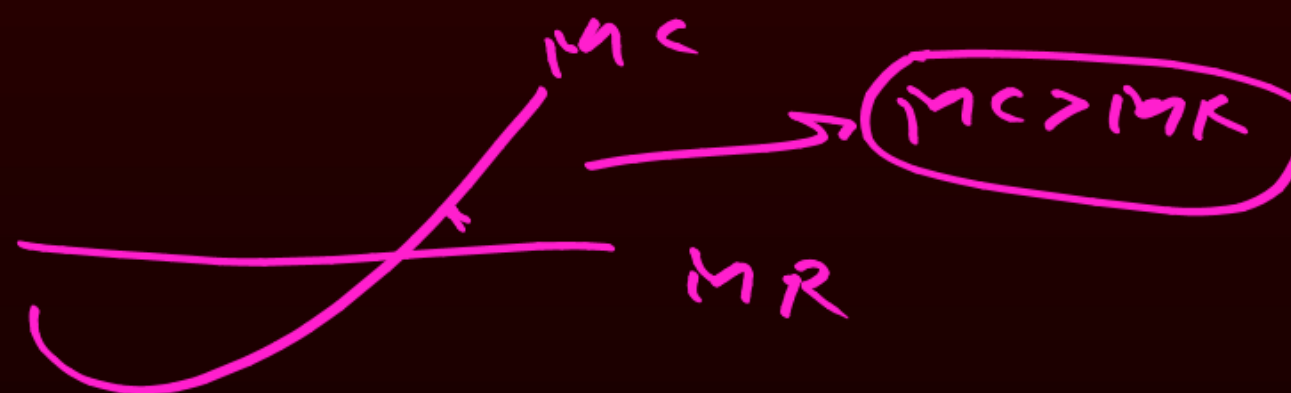
As equilibrium is achieved when $MC = MR$, it means, price is more than MC at the equilibrium level.

MULTIPLE CHOICE QUESTIONS

In case of perfect competition, a firm is in equilibrium when:

- (a) $MC = MR$
- (b) MC cuts MR from below
- (c) MC is rising when it cuts MR
- (d) All of these

Ans. (d) All of these

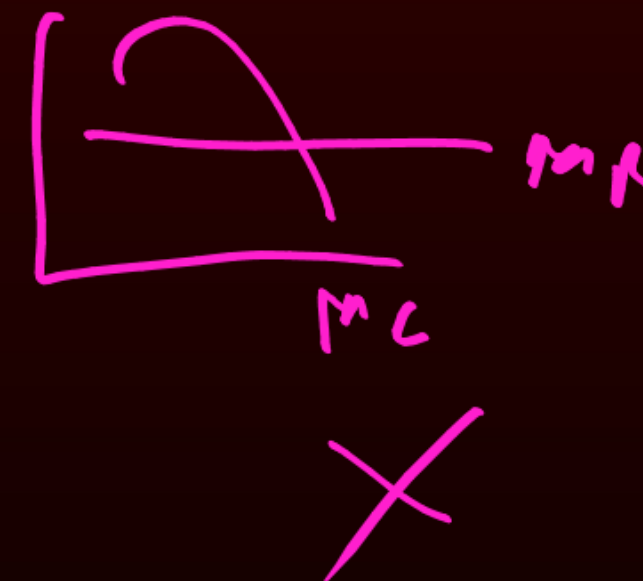


MULTIPLE CHOICE QUESTIONS

If Marginal Cost = MC and Marginal Revenue = MR, then for achieving equilibrium output:

- (a) MC Curve should cut MR Curve from above
- (b) MC Curve should cut MR Curve from below
- (c) MC Curve should not cut MR Curve at all
- (d) MC Curve should be tangent to MR Curve

Ans. (b) MC Curve should cut MR Curve from below



MULTIPLE CHOICE QUESTIONS

Read the following statements carefully and choose the correct alternative from the following:

Statement 1: A producer is said to be in equilibrium when he wishes to expand the output. X

Statement 2: The difference between revenue and cost is termed as profit. TR - TC

Alternatives:

- (a) Both the statements are true.
- (b) Both the statements are false.
- (c) Statement 1 is true and Statement 2 is false.
- ☒ (d) Statement 2 is true and Statement 1 is false.

$$\underline{TR - TC = \text{Profit}}$$

Ans. (d) Statement 2 is true and Statement 1 is false.

MULTIPLE CHOICE QUESTIONS

Read the following statements: Assertion (A) and Reason (R). Choose one of the correct alternatives given below:

Assertion (A): When price remains same at all levels of output, then Price is more than MC at the equilibrium level. *False*

Reason (R): When MC is greater than MR after equilibrium, it means producing more will lead to decline in profits.

Alternatives:

- (a) Both (A) and (R) are True and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are True and (R) is not the correct explanation of (A).
- (c) (A) is True but (R) is False.
- ☒ (d) (A) is False but (R) is True.

Ans. (d) (A) is False but (R) is True.

$$\text{Price} = AR = MR$$

$$MR > MC$$

eq.

$$MR = MC$$

QUESTIONS

Que. What is the general profit maximising condition for a producer (MR and MC approach)?

Ans. (i) $MC = MR$ and ✓

(ii) MC curve cuts the MR curve from below (i.e. MC is rising).

$MC = MR$ ✓

QUESTIONS

Que. What is Producer's Equilibrium?

Ans. Producer's Equilibrium refers to that price and output combination which brings maximum profit to the producer and profit declines as more is produced.

QUESTIONS

Que. Explain producer's equilibrium with the help of a diagram.

Ans. Producer's equilibrium refers to a situation of profit maximisation or cost minimisation. To study producer's equilibrium, two approaches are used

(i) Total Revenue and Total Cost Approach Under this approach, a producer is deemed to be in equilibrium, on the fulfillment of the following two conditions

- (a) The difference between TR and TC is the maximum.
- (b) Total profits are falling after this level of output.

(ii) Marginal Revenue and Marginal Cost Approach Under this approach, a producer attains equilibrium on the fulfillment of the following two conditions

- (a) $MR = MC$. ✓
- (b) MC is rising after the point of equilibrium. ✓

Handwritten note:

$$TR - TC = \text{Profit}$$

Profit Max

QUESTIONS

Que. From the following information about a firm, find the firm's equilibrium output in terms of marginal cost and marginal revenue. Give reasons. Also find profit at this output?

Output (units)	Total Revenue (₹)	Total Cost (₹)
1	6	7
2	12	13
3	18	17
4	24	23
5	30	31

H.W

Ans.

Output (units)	Total Revenue (TR) (₹)	Total Cost (TC) (₹)	Marginal Revenue (MR) (₹)	Marginal Cost (MC) (₹)	Profit (₹) (TR- TC)
1	6	7	6	7	-1
2	12	13	6	6	-1
3	18	17	6	4	1
4	24	23	6	6	1
5	30	31	6	8	-1

QUESTIONS

Que. Explain producer's equilibrium with the help of a diagram.

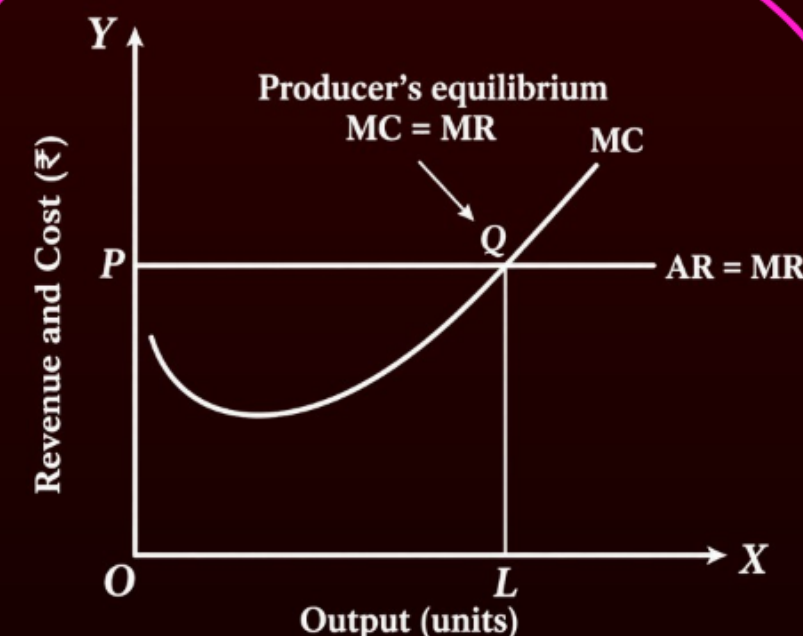
Ans. Producer's equilibrium refers to a situation of profit maximisation or cost minimisation. To study producer's equilibrium, two approaches are used

(i) Total Revenue and Total Cost Approach Under this approach, a producer is deemed to be in equilibrium, on the fulfillment of the following two conditions

- (a) The difference between TR and TC is the maximum.
- (b) Total profits are falling after this level of output.

(ii) Marginal Revenue and Marginal Cost Approach Under this approach, a producer attains equilibrium on the fulfillment of the following two conditions

- (a) $MR = MC$.
- (b) MC is rising after the point of equilibrium.



DOUBT



**THANK
YOU**

Stay Connected

KEEP LEARNING

Reverse
Demand